Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Team 23: Sound Localization Robot
Bi-Weekly Update 5

Samuel Lackey, Sanjana Pasumarthi, Lucas Kettler, Joshue Teele

Sponsor: Dr. Moreira
TA: Muhammed Bin Mughees

Project Scope

Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)

- Keeps **first responders safe** by entering hazardous or toxic areas first.
- **Reduces search and rescue time** by avoiding exploring unnecessary paths
- **Reliable in low-visibility** or high-stress conditions (smoke, debris, noise).

Problem: First Responders lack reliable methods for navigating dangerous environments, where visibility is limited, putting both their safety and the victims at risk.

Our Solution: Building a Sound Localization robot that can hear distress calls, pinpoint their direction, and move toward them autonomously, allowing first responders to focus on saving lives instead of risking theirs.

*A smarter, safer, and faster way to locate victims, all while protecting rescuers and saving more lives.

Project Timeline

Subsystem Designs and Testing (completed 1/29)	Integration of Motor Control, Navigation (completed 2/5)	Integration of Sound Algorithm, Navigation (completed 2/19)	Final Integration (completed 3/3)	System Test (completed 3/10)	Validation (to complete by 3/31)	Demo and Report (to complete by 4/5)

SYSTEM Validation Plan

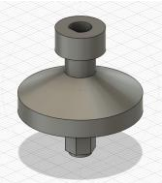
Paragraph	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.1	DOA Angle Accuracy	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance. Run the detection test 5 times	TESTED	Sam
3.2.1.2	DOA With Background Noise	All 5 tests must log a detected angle within an 3.6°	Place speaker playing 450Hz tone at a known angle, and 0.5m distance while separate 550Hz tone is playing. Repeat 5 times	UNTESTED	Sam
3.2.1.3	Obstacle Detection	Verify robot halts movement at the node prior to an object	Isolated test with robot attempting to drive to a goal behind an obstacle	TESTED	Lucas, Joshua, Sanjana
3.2.1.4	Grid Mapping	Verify the robot updates a grid map for the web app	View output of webapp upon completion of path	TESTED	Lucas, Sanjana
3.2.1.5	Webapp	The webapp must show the angle, estimated distance, and control the robot.	Run the program and connect to wifi, trigger a sound, access the webapp, and move.	TESTED	Sam, Sanjana
3.2.1.6	Robot Distance	The robot must finish moving < 0.5 m from the sound source.	Play the sound source, allow the robot to move at different distances and angles.	UNTESTED	Sam
3.2.2.1	Accurate Robot Movement	Robot moves in intended direction determined by program	Series of coordination tests for the robot to move to the sound source	TESTED	Full Team
3.2.2.2	Constant Response rate of robot	Robot has a constant rate of response with +/- 5% of average time	Multiple trials of measuring time it takes for rover to change status	UNTESTED	Full Team
3.2.2.3	Structurally strong Robot	Wheel shaft does not break easily, no unintended movement from motors	Test robot with obstacles and quick movement	TESTED	Sanjana
3.2.3.3.1	Input Voltage(USB)	Ensure voltage at USB port is above 4.75V (minimum needed voltage for linear regulator)	Use multimeter to test output of buck converter	TESTED	Joshua
3.2.3.3.2	Input Voltage(DC jack)	Ensure voltage at DC jack is between 10.8 and 13.2 V (+/-10% of motor voltage)	Use multimeter to test output of DC jack	TESTED	Joshua
N/A	Full System Demo	Robot is able to navigate autonomously around obstacles, reach goal, and display data on app	Multiple observations and trials of robot	TESTED	Full Team

Validation

3.2.2.1	Accurate Robot Movement	Robot moves in intended direction determined by program	Series of coordination tests for the robot to move to the sound source	TESTED	Full Team

Robot moves roughly 1m off every 20m in testing. This can be fine tuned in code

3.2.2.3	Structurally strong Robot	Wheel shaft does not break easily, no unintended movement from motors	Test robot with obstacles and quick movement	TESTED	Sanjana



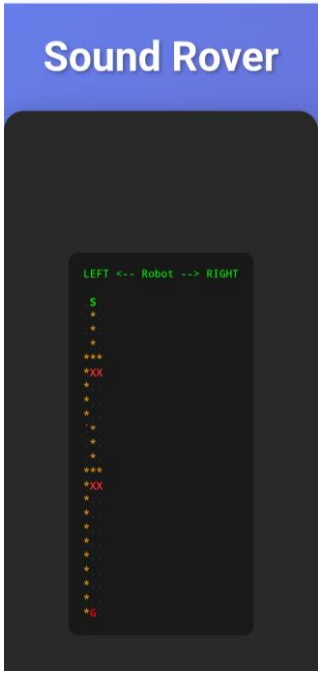
Stronger shafts hold more weight and pressure, naturally fits into wheel for more grip, removing need for adhesive

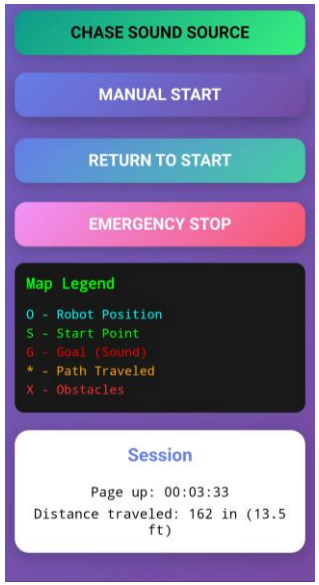
3.2.2.2	Constant Response rate of robot	Robot has a constant rate of response with +/- 5% of average time	Multiple trials of measuring time it takes for rover to change status	NOT TESTED	Full Team

Validation

3.2.1.3	Obstacle Detection	Verify robot halts movement at the node prior to an object	Isolated test with robot trying to drive around obstacle to a goal	TESTED	Lucas, Joshua, Sanjana
3.2.1.4	Grid Mapping	Verify the robot updates a grid map for the web app	View output of webapp upon completion of path	TESTED	Lucas, Sanjana
3.2.1.5	Webapp	The webapp must show the angle, estimate distance, and control the robot.	Run the program and connect to Wi-Fi, access the webapp, and move.	TESTED	Sam, Sanjana

Map updated seamlessly with the live movements, with and without obstacles while showing distance travelled





Validation

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3.2.1.6	Robot Distance	The robot must finish moving < 0.5 m from the sound source.	Play the sound source, allow the robot to move at different distances and angles.	TESTED	Sam

It's difficult to achieve true perfect angle calculation in noisy rooms, but robot will still detect sound source well.

Validation

3.2.3.3.1	Input Voltage(USB)	Ensure voltage at USB port is above 4.75V (minimum needed voltage for linear regulator)	Use multimeter to test output of buck converter	TESTED	Joshua
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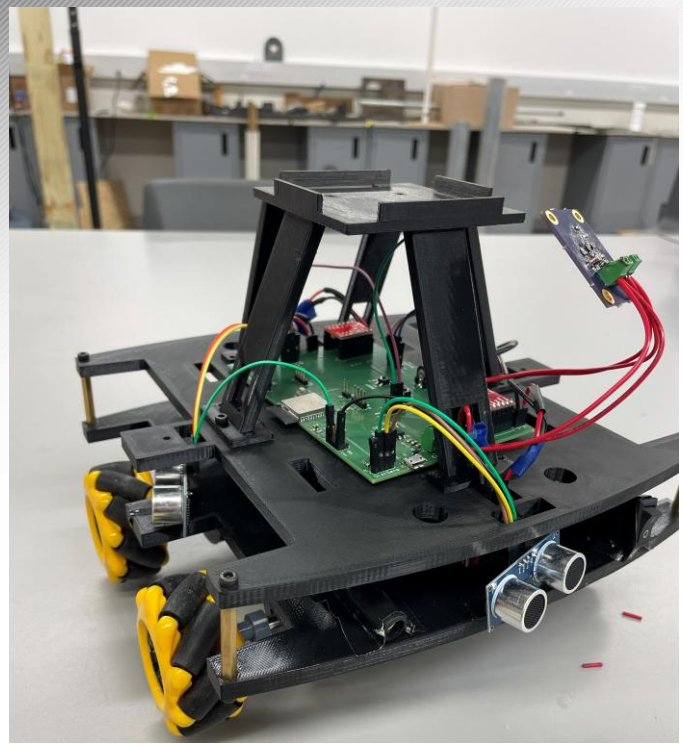
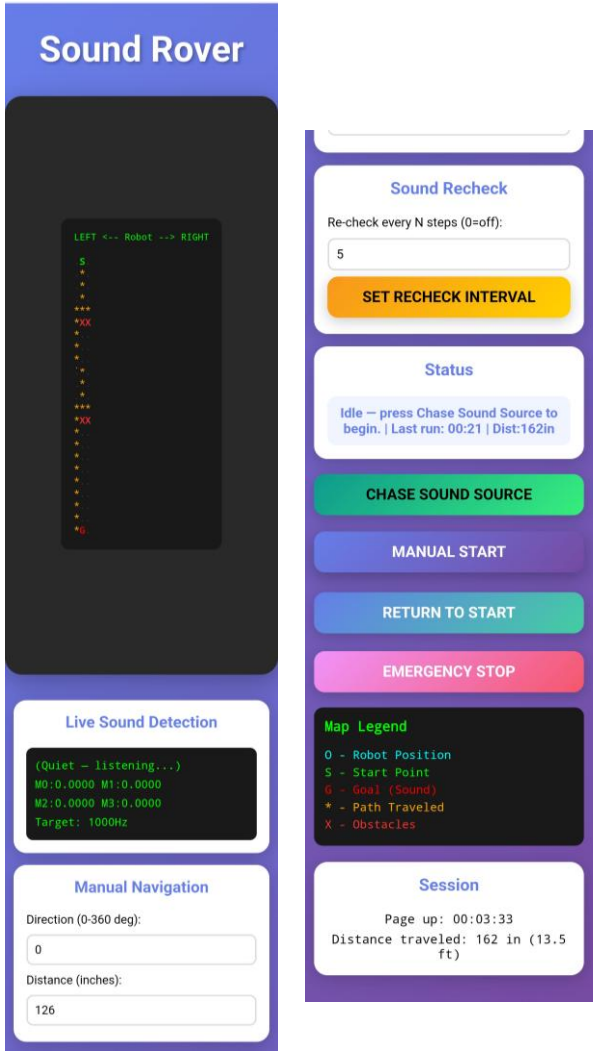
Integrated System Diagram:



Website generated through internal Wi-Fi

User Commands: Emergency Stop, Origin



Fully Integrated System Design:

- Integrated PCB (w/ ESP32) for motor controls, power, distance sensors, and navigation
- Microphone PCB on top layer (not displayed)

User Interface – viewed from any device with Wi-Fi capabilities

Input Signal Software

Samuel Lackey

Accomplishments since last update : 20 Hours of Effort	Ongoing progress/problems and plans until the next presentation
• Fully Integrated and tested new microphone PCB with the robot.	• Continue testing, and improving • I need to work out edge cases and reliability

Mechanical/ Motor Controls Subsystem

Sanjana Pasumarthi

Accomplishments since last update 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated Structural Integrity of chassis design • Completed UI • Validated movement of motors corresponding to their movement	• On-going tests to determine optimal duration of motors for accurate movement • Designing casing to cover set-up

Navigation Software Subsystem

Lucas Kettler

Accomplishments since last update 13 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Validated Grid map live updates • Added a run timer and return to start function • Assisted in tests of varying distances with and without obstacles	• Validate Grid map expansion for demo

Power & Distance Sensor Subsystem

Joshua Teele

Accomplishments since Last Update 12 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Helped other work on fully integrated robot • Changed up wiring on the chassis to make it look neater	• Keep testing and sort out any issues with robot

Execution Plan

Tests	1/15/26	1/22/26	1/29/26	2/5/26	2/12/26	2/19/26	2/26/26	3/5/26	3/12/26	3/19/26	3/26/26 - DONE
Motor Controls Subsystem ~ Sanjana Pasumarthi											
Design and 3D Print Proper Wheel Shaft											
Design integrated PCB w/ Josh for power and motor controls											
Design and 3D Print Top Layer											
Integrate Sound Rover Firmware with Navigation & Sound Software & Test Rover											
Navigation Software Subsystem ~ Lucas Kettler											
Integrate with Input Signal Code											
Integrate with Sensor Code											
Integrate with Motor Control Code											
Adapt Grid map for use with web app											
Adapt code from simulation to Real test code											
Power & Distance Sensor Subsystem ~ Joshua Teele											
Design MCU Board											
Solder PCBs											
Integrate Sensor Code with Lucas											
Input Signal Subsystem ~ Samuel Lackey											
Design Microphone Array PCB											
Integrate Input Signal Software with Navigation											
Finalize all Software Integration											
Solder and Connect PCB to Team system											
Validation Angle Accuracy Tests											

Thank You,
Any Questions?

Sound Localization Robot 5 – Team 23

Lucas Kettler, Samuel Lackey, Joshua Teele,
Sanjana Pasumarthi